

VLE for water + ethanol + 1-octanol mixtures. Experimental measurements and correlations

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Abstract

Molar excess Gibbs free energies (G^E/RT) for the ternary system water + ethanol + 1-octanol were evaluated from the corresponding isobaric (101.32 kPa) vapour–liquid equilibrium data. The G^E/RT –composition data were then correlated by means of the Redlich–Kister polynomial and the NRTL and UNIQUAC equations, using an optimized value of 5.50 for the q' UNIQUAC area parameter of 1-octanol. The experimental data were compared with data predicted using the ASOG, UNIFAC and UNIFAC–Lyngby group contribution methods.

Keywords: Vapour–liquid; Data; Water; Ethanol; 1-octanol; Correlation

1. Introduction

Thermodynamic description of a liquid mixture is based on an equation relating its molar excess Gibbs free energy (G^E/RT , in its dimensionless form) to its composition and, if possible, to temperature and pressure. Such expressions are of particular interest because activity coefficients for mixture components can be derived from them.

In this work, experimental G^E/RT –composition data for the ternary system water + ethanol + 1-octanol were correlated using models selected from among those available in the literature for mixtures of components with widely differing volatilities. The G^E/RT data were obtained using the well-known equation

$$G^E/RT = \sum_i x_i \ln \gamma_i \quad (1)$$

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Table 1

Densities (d), refractive indices (n_D), boiling points (T) and Antoine's constants (A, B, C) for the compounds used

Compound	$d(298.15\text{ K})$ g cm^{-3}		$n_D(298.15\text{ K})$		$T(101.32\text{ kPa})/\text{K}$		A	B	C
	Exp.	Lit. ^a	Exp.	Lit. ^a	Exp.	Lit. ^a	(with P_i^S/kPa and $T/^\circ\text{C}$)		
Water	0.9970	0.99705	1.3324	1.33250	373.15	373.15	7.23255	1750.286	235.000 ^b
Ethanol	0.7851	0.78504	1.3592	1.35941	351.56	351.443	7.16879	1552.601	222.419 ^a
1-Octanol	0.8217	0.82209	1.4275	1.42750	467.85	468.306	5.88511	1264.322	130.73 ^a

^a Riddick et al. (1986).^b Hirata et al. (1975).

where γ_i is the activity coefficient. The required γ_i were evaluated from isobaric (101.32 kPa) vapour–liquid equilibrium (VLE) data for the ternary system by means of the equation

$$\gamma_i = \frac{y_i \phi_i P}{x_i P_i^S \phi_i^S \exp\left(\frac{V_i^L (P - P_i^S)}{RT}\right)} \quad (2)$$

in which x_i and y_i are the mole fractions of component i in the liquid and vapour phases, respectively, V_i^L is the molar volume of component i in the liquid phase as given by the correlation of Yen and Woods (1966), ϕ_i and ϕ_i^S are its fugacity coefficients at the equilibrium pressure P and at saturation, respectively, as calculated by the method of Hayden and O'Connell (1975), and P_i^S is its saturated vapour pressure as calculated from Antoine's equation using parameters (see Table 1) taken from Riddick et al. (1986) and Hirata et al. (1975).

The correlation models used were:

- the Redlich and Kister (1948) polynomial, which for a ternary system has the form

$$\begin{aligned} G^E/RT &= (G^E/RT)_{21} + (G^E/RT)_{23} + (G^E/RT)_{31} \\ &+ x_1 x_2 x_3 (A + B(x_2 - x_1) + C(x_2 - x_3) + D(x_3 - x_1) + \dots) \end{aligned} \quad (3)$$

where the $(G^E/RT)_{ij}$ terms are obtained by fitting the G^E/RT –composition data for the binary sub-systems of the ternary system to the equation

$$(G^E/RT)_{ij} = x_i x_j (a + b(x_i - x_j) + c(x_i - x_j)^2 + \dots) \quad (4)$$

- the NRTL equation (Renon and Prausnitz, 1968)

$$G^E/RT = \sum_i x_i \frac{\sum_j \tau_{ji} G_{ji} x_j}{\sum_k G_{ki} x_k} \quad (5)$$

where

$$G_{ji} = \exp(-\alpha_{ji} \tau_{ji}) = \exp\left(-\alpha_{ji} \frac{\Delta g_{ji}}{T}\right) \quad (6)$$

- and the UNIQUAC equation (Abrams and Prausnitz, 1975; Anderson and Prausnitz, 1978)

$$G^E/RT = \sum_i x_i \ln \frac{\Phi_i}{x_i} + \frac{z}{2} \sum_i q_i x_i \ln \frac{\theta_i}{\Phi_i} - \sum_i q'_i x_i \ln \left(\sum_j \theta'_j \tau_{ji} \right) \quad (7)$$

where

$$\tau_{ji} = \exp \left(\frac{-\Delta u_{ji}}{T} \right) \quad (8)$$

An alternative approach is to predict G^E/RT values. In this work we examined prediction using the ASOG (Kojima and Tochigi, 1979), UNIFAC (Fredenslund et al., 1977a,b) and UNIFAC–Lyngby (Larsen et al., 1987) group contribution methods.

2. Experimental

Water was purified using a Milli-Q plus system. Ethanol (Merck) and 1-octanol (Aldrich) both had nominal purities of > 99.5 mass%. Their water contents (0.08 mass% for ethanol and 0.02 mass% for 1-octanol) were determined using a Metrohm 737 KF Coulometer. Table 1 includes the measured densities, refractive indices and boiling points of these chemicals, together with published values for these parameters (Riddick et al., 1986).

2.1. Apparatus and procedure

VLE measurements were performed in a Labodest apparatus that recycles both liquid and vapour phases (Fischer Labor und Verfahrenstechnik, Germany). This was equipped with a Fischer digital manometer and a Heraeus QuaT100 quartz thermometer that measured to within ± 0.01 kPa and ± 0.02 K respectively. The inert atmosphere in the apparatus was argon, which was maintained at a constant pressure of 101.32 kPa.

Vapour and liquid phase compositions were determined by densimetry and refractometry using previously published data for the composition dependence of the densities and refractive indices of the systems studied (Arce et al., 1993). The maximum deviation in composition was ± 0.001 mole fraction, as confirmed by comparison of selected results with those for samples prepared by weighing.

3. Results and data analysis

The isobaric vapour–liquid equilibrium data at 101.32 kPa for the binary subsystem ethanol + 1-octanol have been published previously (Arce et al., 1995). The VLE data and liquid phase activity coefficients (γ_i) for the sub-system ethanol + water are compared with those reported by other authors (Paul, 1976; Zemp and Francesconi, 1992 and Kurihara et al., 1993) in Fig. 1. The experimental values for x , y and T together with γ_i and derived G^E/RT values for this sub-system are listed in Table 2. Fredenslund's test (Fredenslund et al., 1977b) confirmed the VLE data to be thermodynamically consistent. The VLE and G^E/RT data for the miscible region of the ternary

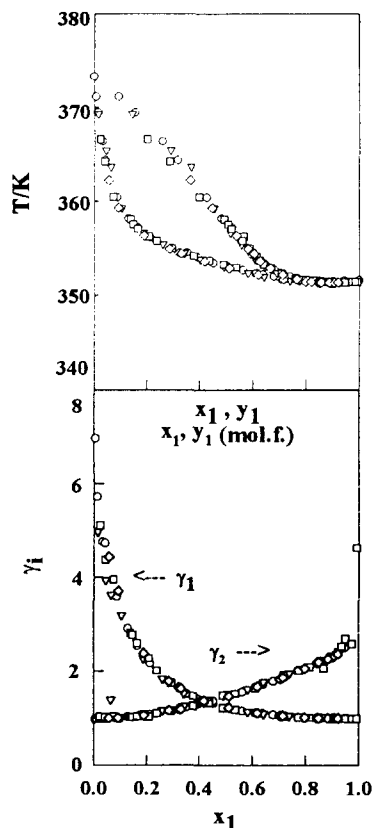


Fig. 1. Comparison of experimental vapour–liquid equilibrium data and activity coefficients for the ethanol(1) + water(2) system at 101.32 kPa: $\circ$, this work; ∇ , Paul (1976), $\square$, Zemp and Francesconi (1992), $\diamond$, Kurihara et al. (1993).

system water(1) + ethanol(2) + 1-octanol(3) are listed in Table 3. Fig. 2 shows the temperature–composition diagram with the temperature isolines for the liquid phase of the ternary system and the contour of the immiscible region (Arce et al., 1994).

The G^E/RT –composition data for the ternary system were first fitted by the Redlich and Kister (1948) polynomial, which, owing to its algebraic flexibility, can be adequately fitted to any excess function. For example, in contrast to Legendre Polynomials (previously fitted to G^E/RT data for the system ethanol + 1-octanol; Arce et al., 1995), the Redlich–Kister polynomial adequately reflects the negative G^E/RT obtained for 1-octanol-rich mixtures. The term $(G^E/RT)_{31}$, for the partially miscible binary subsystem water + 1-octanol, was omitted from the polynomial. Curve-fitting, in the first instance to the data for the binary subsystems, was by least-squares regression, applying Fisher's *F*-test (Akhnazarova and Kafarov, 1982) to minimize the number of parameters in the final expression. The resulting coefficients and root-mean-square deviations between calculated and experimental G^E/RT are listed in Table 4, and the G^E/RT –composition surface for the ternary system is plotted in Fig. 3.

The G^E/RT –composition data for the ternary system were subsequently correlated using NRTL

Table 2

Experimental vapour–liquid equilibrium data, activity coefficients and excess Gibbs free energies (G^E/RT) for the ethanol(1)+water(2) system at 101.32 kPa

x_1	y_1	T/K	γ_1	γ_2	G^E/RT
0.0000	0.0000	373.15	–	1.0000	0.0000
0.0317	0.2573	366.29	4.7509	0.9781	0.0279
0.0424	0.3192	364.32	4.7170	0.9754	0.0419
0.0863	0.4289	360.33	3.5852	0.9974	0.1078
0.1300	0.4830	358.10	2.9050	1.0335	0.1673
0.1666	0.5221	357.16	2.5357	1.0346	0.1833
0.2137	0.5511	356.07	2.1719	1.0750	0.2226
0.2930	0.5847	354.97	1.7506	1.1553	0.2661
0.3525	0.6031	354.50	1.5273	1.2284	0.2825
0.3950	0.6150	353.99	1.4167	1.3014	0.2970
0.4531	0.6412	353.40	1.3164	1.3739	0.2983
0.5060	0.6530	353.01	1.2182	1.4942	0.2983
0.5629	0.6833	352.64	1.1619	1.5646	0.2801
0.6142	0.7056	352.33	1.1126	1.6687	0.2631
0.6395	0.7182	352.15	1.0951	1.7220	0.2540
0.6794	0.7410	351.95	1.0715	1.7943	0.2344
0.7240	0.7683	351.77	1.0497	1.8786	0.2091
0.7740	0.7973	351.57	1.0267	2.0238	0.1797
0.8436	0.8505	351.48	1.0082	2.1658	0.1277
0.8612	0.8649	351.44	1.0058	2.2092	0.1150
0.9020	0.9016	351.42	1.0018	2.2816	0.0824
0.9464	0.9409	351.48	0.9941	2.5004	0.0435
1.0000	1.0000	351.56	1.0000	–	0.0000

and UNIQUAC equations, in both cases by means of the Levenberg–Marquardt non-linear regression method. The objective function minimized (F being the r.m.s. deviation in G^E/RT) was

$$F = \left(\frac{1}{N} \sum_{i=1}^n \left((G^E/RT)_{i,\text{exp}} - (G^E/RT)_{i,\text{calc}} \right)^2 \right)^{1/2} \quad (9)$$

The procedure was the same for both models. (i) The G^E/RT –composition data for the ternary system were correlated, and binary interaction parameters and r.m.s. deviations between calculated and experimental G^E/RT were obtained. (ii) These parameters and r.m.s. deviations were compared with those obtained for correlation of the G^E/RT –composition data for the miscible binary sub-systems. The results for each model are discussed below.

4. NRTL equation

A value of 0.1 for the NRTL non-randomness factor (α) proved adequate for correlation of the G^E/RT data for the miscible binary sub-systems, and was maintained for correlation of these data for the ternary system. The G^E/RT –composition data for the ternary system were well correlated using

Table 3

Experimental vapour–liquid equilibrium data, activity coefficients and molar excess Gibbs free energies (G^E/RT) for the water(1)+ethanol(2)+1-octanol(3) system at 101.32 kPa

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.0229	0.9616	0.0276	0.9718	351.82	2.7003	0.9940	5.2234	0.0421
0.1028	0.8749	0.1097	0.8900	351.95	2.3763	0.9959	1.7698	0.0981
0.2206	0.7586	0.2099	0.7899	352.18	2.0973	1.0107	1.2502	0.1761
0.0453	0.9234	0.0539	0.9454	352.23	2.6215	0.9917	2.8840	0.0691
0.1468	0.8200	0.1559	0.8435	352.23	2.3374	0.9965	2.3381	0.1499
0.0415	0.9226	0.0506	0.9484	352.31	2.6779	0.9926	3.5729	0.0798
0.0433	0.9077	0.0553	0.9436	352.64	2.7680	0.9913	2.8191	0.0870
0.1395	0.7925	0.1646	0.8344	353.05	2.5130	0.9887	1.8047	0.1597
0.0975	0.8336	0.1162	0.8829	353.06	2.5384	0.9941	1.5996	0.1183
0.2077	0.7214	0.2298	0.7693	353.31	2.3307	0.9917	1.5353	0.2001
0.0469	0.8687	0.0582	0.9404	353.55	2.5936	0.9974	1.9646	0.0994
0.4714	0.5109	0.3508	0.6485	353.61	1.5475	1.1676	4.7108	0.3124
0.2260	0.6876	0.2589	0.7397	353.73	2.3727	0.9848	1.9093	0.2406
0.4645	0.5001	0.3630	0.6362	354.09	1.5943	1.1492	2.6115	0.3202
0.1310	0.7558	0.1709	0.8277	354.09	2.6658	0.9888	1.4200	0.1596
0.2740	0.6272	0.2962	0.7029	354.27	2.1909	1.0054	1.0383	0.2220
0.0901	0.7849	0.1219	0.8765	354.41	2.7311	0.9961	1.4378	0.1329
0.0379	0.8348	0.0588	0.9393	354.62	3.1079	0.9957	1.6511	0.1032
0.4542	0.4860	0.3782	0.6209	354.69	1.6588	1.1285	1.6748	0.3195
0.1888	0.6740	0.2480	0.7502	354.73	2.6153	0.9813	1.4498	0.2197
0.1251	0.7256	0.1694	0.8289	354.92	2.6777	0.9998	1.2401	0.1552
0.6075	0.3761	0.4100	0.5899	354.98	1.3289	1.3707	0.6670	0.2847
0.1188	0.7178	0.1783	0.8202	355.27	2.9270	0.9871	0.9782	0.1146
0.4366	0.4717	0.3955	0.6034	355.42	1.7532	1.0996	1.2752	0.3122
0.1782	0.6428	0.2573	0.7408	355.70	2.7669	0.9799	1.1035	0.1859
0.1029	0.7240	0.1389	0.8590	355.71	2.5884	1.0082	1.2557	0.1432
0.5963	0.3614	0.4343	0.5639	355.79	1.3890	1.3231	4.4247	0.3600
0.0361	0.7806	0.0616	0.9362	356.18	3.2148	1.0014	1.2034	0.0772
0.4189	0.4488	0.4145	0.5834	356.20	1.8572	1.0855	1.6074	0.3590
0.5914	0.3413	0.4577	0.5404	356.56	1.4320	1.3050	2.7992	0.3725
0.1090	0.6702	0.1855	0.8129	356.68	3.1405	0.9943	0.7069	0.0443
0.3990	0.4219	0.4381	0.5587	357.16	1.9849	1.0674	1.7051	0.3966
0.5798	0.3071	0.4951	0.5022	357.85	1.5024	1.2855	2.1872	0.4017
0.5178	0.3310	0.4939	0.5030	358.04	1.6660	1.1862	1.8564	0.4144
0.1526	0.5670	0.2737	0.7235	358.13	3.1264	0.9919	0.8928	0.1375
0.3702	0.3945	0.4590	0.5372	358.23	2.1502	1.0554	1.4435	0.3910
0.1018	0.6239	0.1868	0.8111	358.23	3.1885	1.0067	0.6785	0.0158
0.0325	0.7100	0.0608	0.9358	358.34	3.2409	1.0163	1.1579	0.0875
0.3087	0.4285	0.4279	0.5694	358.36	2.3922	1.0249	0.9101	0.2550
0.4259	0.3479	0.4907	0.5062	358.63	1.9671	1.1115	1.1965	0.3655
0.2268	0.4796	0.3709	0.6261	358.64	2.7931	0.9964	0.8881	0.1964
0.4976	0.3004	0.5213	0.4756	358.97	1.7652	1.1948	1.3136	0.3913
0.3817	0.3556	0.4895	0.5070	359.10	2.1504	1.0707	1.1301	0.3487
0.3287	0.3849	0.4640	0.5339	359.11	2.3664	1.0412	0.6212	0.1623
0.3422	0.3635	0.4817	0.5144	359.50	2.3246	1.0473	1.0966	0.3326
0.2863	0.3968	0.4461	0.5507	359.66	2.5579	1.0210	0.8267	0.2168
0.4338	0.3004	0.5312	0.4645	359.77	2.0009	1.1335	1.3191	0.4121

Table 3 (continued)

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.2294	0.4363	0.3900	0.6065	359.84	2.7727	1.0158	0.8462	0.1850
0.1391	0.5115	0.2913	0.7055	360.12	3.3815	0.9974	0.7256	0.0561
0.3979	0.2990	0.5348	0.4608	360.30	2.1522	1.1083	1.1464	0.3771
0.3098	0.3536	0.4823	0.5138	360.40	2.4840	1.0409	0.9079	0.2636
0.0263	0.6494	0.0604	0.9356	360.43	3.6722	1.0298	0.9526	0.0375
0.0905	0.5621	0.1864	0.8104	360.57	3.2720	1.0255	0.7081	0.0015
0.3373	0.3135	0.5195	0.4752	360.85	2.4154	1.0686	1.1587	0.3697
0.3042	0.3374	0.4971	0.4982	360.93	2.5552	1.0378	0.9957	0.2964
0.2049	0.4126	0.3857	0.6106	361.18	2.9175	1.0304	0.7211	0.1067
0.2586	0.3574	0.4646	0.5316	361.40	2.7602	1.0278	0.7298	0.1514
0.1263	0.4897	0.2766	0.7197	361.48	3.3586	1.0119	0.7032	0.0236
0.0735	0.5534	0.1596	0.8364	361.73	3.3020	1.0311	0.7682	0.0064
0.2687	0.3331	0.4903	0.5052	361.85	2.7557	1.0314	0.8120	0.1997
0.1288	0.4524	0.3183	0.6772	362.32	3.6706	1.0004	0.7468	0.0454
0.1089	0.4855	0.2399	0.7549	362.50	3.2520	1.0322	0.8795	0.0918
0.2991	0.2854	0.5396	0.4540	362.61	2.6471	1.0531	1.0593	0.3299
0.2658	0.2997	0.5161	0.4777	363.00	2.8080	1.0406	0.9583	0.2678
0.3533	0.2321	0.6063	0.3874	363.04	2.4770	1.0887	1.0210	0.3488
0.2310	0.3233	0.4773	0.5174	363.29	2.9564	1.0340	0.7842	0.1528
0.1798	0.3737	0.3923	0.6032	363.37	3.1141	1.0395	0.6598	0.0331
0.0663	0.5178	0.1546	0.8402	363.39	3.3317	1.0436	0.8120	0.0153
0.2397	0.3030	0.4932	0.5014	363.84	2.8840	1.0486	0.7543	0.1394
0.1175	0.4187	0.3199	0.6745	364.22	3.7665	1.0066	0.7505	0.0254
0.0822	0.4664	0.2050	0.7895	364.45	3.4238	1.0489	0.7448	−0.0096
0.0545	0.4983	0.1349	0.8592	364.89	3.3452	1.0519	0.7847	−0.0174
0.1096	0.3990	0.3200	0.6734	365.25	3.8880	1.0172	0.7863	0.0375
0.2477	0.2473	0.5564	0.4362	365.39	2.9717	1.0589	0.8569	0.2060
0.0880	0.4261	0.2323	0.7607	365.84	3.4418	1.0538	0.8132	0.0306
0.1951	0.2798	0.4887	0.5046	366.46	3.1870	1.0428	0.7003	0.0508
0.0592	0.4445	0.1641	0.8286	366.91	3.4770	1.0602	0.7796	−0.0238
0.1738	0.2733	0.4741	0.5183	368.01	3.2802	1.0395	0.6905	0.0123
0.0858	0.3842	0.2149	0.7768	368.74	2.9382	1.0800	0.7496	−0.0307
0.2508	0.1171	0.7268	0.2633	370.63	3.1677	1.1290	0.6847	0.0640
0.1857	0.1977	0.5637	0.4252	370.77	3.3036	1.0737	0.7771	0.0805
0.1423	0.2353	0.4630	0.5256	372.15	3.3732	1.0643	0.7311	−0.0073
0.1786	0.1592	0.6067	0.3795	373.21	3.3903	1.0972	0.7884	0.0753
0.2163	0.1121	0.7112	0.2759	373.29	3.2711	1.1305	0.7257	0.0548
0.0497	0.3452	0.1708	0.8146	373.33	3.4244	1.0803	0.8965	0.0217
0.1596	0.1310	0.6275	0.3554	376.29	3.5245	1.1288	0.7742	0.0354
0.1063	0.1882	0.4516	0.5287	379.23	3.4468	1.0627	0.7664	−0.0448
0.1227	0.1452	0.5525	0.4256	379.96	3.5625	1.0838	0.7928	−0.0024
0.0570	0.2227	0.2744	0.6998	381.95	3.5684	1.0904	0.8527	−0.0230
0.1104	0.1194	0.5832	0.3895	383.97	3.6559	1.0643	0.7689	−0.0519
0.0874	0.1596	0.4273	0.5393	387.11	3.0563	1.0011	0.8227	−0.0492
0.0617	0.0860	0.5030	0.4429	396.31	3.8140	1.1658	0.7700	−0.1270
0.0441	0.1174	0.3232	0.6146	397.51	3.3089	1.1450	0.8499	−0.0677
0.0489	0.0989	0.3935	0.5353	401.20	3.2501	1.0686	0.8162	−0.1089

Table 3 (continued)

x_1	x_2	y_1	y_2	T/K	γ_1	γ_2	γ_3	G^E/RT
0.0233	0.0912	0.2158	0.6821	408.89	2.9944	1.2020	0.8149	−0.1390
0.0299	0.0450	0.4301	0.4273	414.98	3.9179	1.3067	0.8584	−0.0884
0.0065	0.0268	0.2304	0.5188	428.22	6.8147	1.9439	0.8837	−0.0892
0.0143	0.0216	0.3037	0.3583	435.43	3.4114	1.4206	0.9332	−0.0415
0.0123	0.0185	0.1742	0.2605	449.87	1.6271	0.9007	0.9809	−0.0146

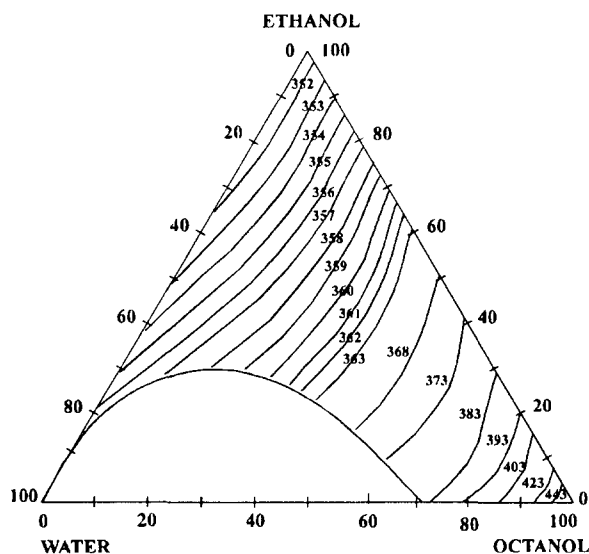


Fig. 2. Temperature–composition diagram with temperature isolines for the liquid phase of the system water + ethanol + 1-octanol.

Table 4

Correlation of the G^E/RT –composition data for the water+ethanol+1-octanol system using the Redlich–Kister polynomial. Coefficients and root-mean-square deviations between calculated and experimental G^E/RT

System	a	b	c	d	r.m.s.
Ethanol + water	1.173	0.2703			0.005
Ethanol + 1-octanol	0.0981	0.4069	0.3697	0.5550	0.008
	A	B	C	D	r.m.s.
Water + ethanol + 1-octanol	4.699	−7.846	0.9005	−6.946	0.033

the NRTL equation, which yielded moderately small r.m.s. deviations in the G^E/RT for both this system and its binary subsystems (Table 5).

The fact that binary interaction parameters are generally applicable to multi-component mixtures led us to set these parameters for the miscible components of the ternary system to the values obtained for their binary mixtures. Correlation of the data for the ternary mixture is thus simplified to calculation of the remaining parameters, Δg_{13} and Δg_{31} . Indeed, the binary interaction parameters

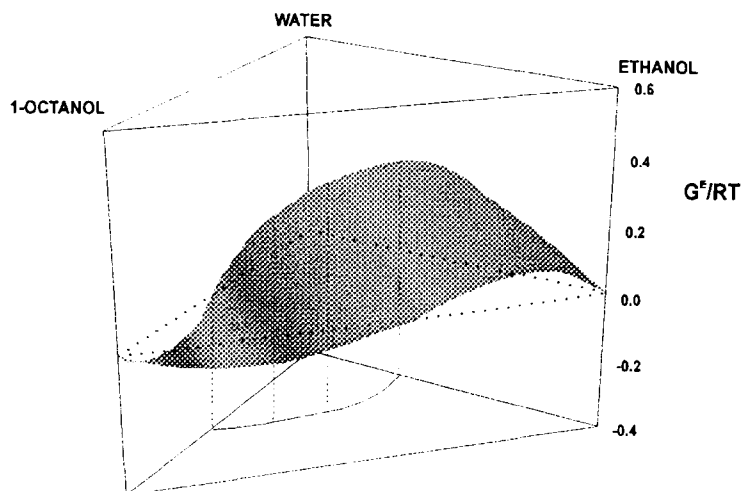


Fig. 3. G^E/RT -composition surface obtained by fitting the Redlich–Kister polynomial to the data for the water + ethanol + 1-octanol system.

Table 5

Correlation of the G^E/RT -composition data for the water(1)+ethanol(2)+1-octanol(3) system using the NRTL and UNIQUAC equations. Binary interaction parameters and root-mean-square deviations between calculated and experimental G^E/RT

Model							r.m.s. deviation		
							Ternary system	Water + ethanol	Ethanol + 1-octanol
NRTL	$\Delta g_{12}/K$	$\Delta g_{21}/K$	$\Delta g_{13}/K$	$\Delta g_{31}/K$	$\Delta g_{23}/K$	$\Delta g_{32}/K$	0.041	0.013	0.026
	1144.2	-530.63	4154.8	-1300.2	2811.4	-1568.3			
UNIQUAC	$\Delta u_{12}/K$	$\Delta u_{21}/K$	$\Delta u_{13}/K$	$\Delta u_{31}/K$	$\Delta u_{23}/K$	$\Delta u_{32}/K$	0.050	0.010	0.031
	248.64	135.89	639.24	23.484	551.46	-257.49			

Table 6

Simplified correlation of the G^E/T -composition data for the water(1)+ethanol(2)+1-octanol(3) system using NRTL and UNIQUAC equations. Binary interaction parameters for these correlations for the miscible binary sub-systems (1+2) and (2+3), binary interaction parameters calculated for the partially miscible components (1 and 3), and root-mean-square deviations between calculated and experimental G^E/RT

Model							r.m.s. deviation		
							Ternary system	Water + ethanol	Ethanol + 1-octanol
NRTL	$\Delta g_{12}/K$	$\Delta g_{21}/K$	$\Delta g_{13}/K$	$\Delta g_{31}/K$	$\Delta g_{23}/K$	$\Delta g_{32}/K$	0.045	0.006	0.013
	1014.8	-431.24	3885.8	-1364.4	2816.3	-1530.8			
UNIQUAC	$\Delta u_{12}/K$	$\Delta u_{21}/K$	$\Delta u_{13}/K$	$\Delta u_{31}/K$	$\Delta u_{23}/K$	$\Delta u_{32}/K$	0.053	0.008	0.022
	342.14	77.177	696.22	-21.225	445.30	-202.96			

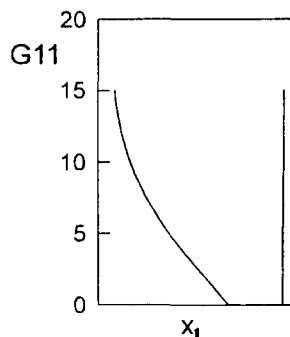


Fig. 4. G_{11} -composition curve derived for the water(1) + 1-octanol(2) system from G^E/RT -composition data for the water + ethanol + 1-octanol system that was correlated using the NRTL equation and the simplified approach.

for the ternary system (Table 5) and the miscible binary subsystems (Table 6) were very similar, suggesting that the simplified approach to correlation might be feasible. In fact, this approach produced considerably decreased r.m.s. deviations in G^E/RT for the binary sub-systems, while only slightly increasing the deviation in G^E/RT for the ternary system (Table 6).

A simple verification of the physical significance of the NRTL binary interaction parameters for the subsystem water + 1-octanol was carried out for the simplified approach (parameters from either of the NRTL correlations serve, since there are only slight variations between them). Firstly, the second derivative of the G^E/RT -composition expression was obtained and G_{11} was calculated in accordance with the equation

$$G_{11} = \frac{\partial^2(G^M/RT)}{\partial x_1^2} = \frac{1}{x_1 x_2} + \frac{\partial^2(G^E/RT)}{\partial x_1^2} \quad (10)$$

Then, bearing in mind that for a thermodynamically stable system $G_{11} > 0$, and for a thermodynamically unstable, i.e. only partially miscible, system $G_{11} < 0$, the ability of these G_{11} values to predict the immiscibility of this binary subsystem was ascertained. Fig. 4 shows the resulting G_{11} -composition curve. The negative values of G_{11} confirm the validity of the binary interaction parameters.

5. UNIQUAC equation

The UNIQUAC area and volume structural parameters (q and r , respectively) were calculated from group contribution data (Gmehling et al., 1982). The q' parameters for water and ethanol were taken from the work of Anderson and Prausnitz (1978), and q' for 1-octanol was obtained from the G^E/RT -composition correlation for the binary subsystem ethanol + 1-octanol using the DSC-Powell optimization method. The optimum value obtained ($q' = 5.50$) proved adequate for correlation of the ternary G^E/RT -composition data.

The binary interaction parameters and r.m.s. deviations obtained by correlation of the data for the ternary system are included in Table 5, and these data for the correlation in which the interaction parameters for the miscible sub-systems were fixed are included in Table 6. All the r.m.s. deviations

Table 7

Root-mean-square deviations between experimental G^E/RT for the water+ethanol+1-octanol system and G^E/RT predicted using the indicated group contribution methods

Method	r.m.s. deviation
ASOG	0.145
UNIFAC	0.136
UNIFAC–Lyngby	0.163

are larger than the corresponding values obtained using the NRTL equation, and there is little similarity between the binary interaction parameters calculated for the water(1) + 1-octanol(3) sub-system using the two approaches.

6. Prediction

The ASOG, UNIFAC (using parameters from Gmehling et al., 1982) and UNIFAC–Lyngby group contribution methods were used to predict the G^E/RT data for the ternary system water(1) + ethanol(2) + 1-octanol(3). None of these methods predicted the negative values observed for this excess function of mixtures rich in 1-octanol (Fig. 3). Otherwise, the best predictions were obtained by the UNIFAC method, although the r.m.s. deviations between the predicted and experimental values of G^E/RT were in all cases very large (Table 7).

7. Conclusions

This paper reports vapour–liquid equilibrium data for water + ethanol + 1-octanol mixtures at 101.32 kPa.

Correlation of the G^E/RT –composition data for the ternary system water + ethanol + 1-octanol demonstrates once again that the NRTL equation is the most suitable correlation model for aqueous systems. By setting the binary interaction parameters for the miscible components of the ternary system to the values obtained by NRTL correlation of the G^E/RT –composition data for their binary mixtures, correlation was reduced to calculation of only two parameters. This simplified approach afforded considerably smaller r.m.s. deviations between calculated and experimental G^E/RT for the miscible binary sub-systems at the expense of only a small increase in this deviation for the ternary system.

As regards prediction of the G^E/RT data using group contribution methods, the UNIFAC method afforded the best predictions. Nonetheless, the r.m.s. deviation between the predicted and experimental values was still excessively large.

8. List of symbols

$a, b, c \dots$ adjustable coefficients in Eq. (4)
 $A, B, C, D \dots$ adjustable coefficients in Eq. (3)

d	density (g cm^{-3})
F	root-mean-square deviation in G^E/RT
g	NRTL binary interaction parameter (K)
G	molar Gibbs free energy (J mol^{-1})
G	parameter in the NRTL equation
G_{11}	second derivative, with respect to mole fraction of component 1, of the molar Gibbs free energy of mixing
n_D	refractive index
N	number of experimental points
P	pressure (kPa)
q, q'	UNIQUAC area parameters
r	UNIQUAC volume parameter
r.m.s.	root-mean-square
R	gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
T	temperature (K)
u	UNIQUAC binary interaction parameter (K)
v	molar volume (l mol^{-1})
x	mole fraction in the liquid phase
y	mole fraction in the vapour phase
z	coordination number

8.1. Greek letters

α	NRTL non-randomness parameter
ϕ	fugacity coefficient
Φ	UNIQUAC volume fraction
γ	activity coefficient
θ, θ'	UNIQUAC area fraction
τ	NRTL and UNIQUAC correlation coefficient

8.2. Superscripts

E	excess
L	liquid
M	mixing
S	saturation

8.3. Subscripts

i, j	components
Calc	calculated value
Exp	experimental value

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